Analyses

March 19, 2025

[132]:
```
from pathlib import Path
import scipy.io as sio
import pandas as pd
import numpy as np
import matplotlib as plt
import pursuit_functions
import pyarrow as pa
import seaborn as sns
import matplotlib.pyplot as plt
from scipy.stats import zscore
from sklearn.decomposition import PCA
from sklearn.preprocessing import StandardScaler

pd.set_option('display.max_columns', 100)  # Show more columns (default is 20)

%load_ext autoreload
%autoreload 2
```

```
The autoreload extension is already loaded. To reload it, use:
  %reload_ext autoreload
```

[55]:
```
data_dir = Path("/Users/may/pursuit")
all_regions_data = pd.read_parquet("/Users/may/pursuit/ca1_ca3_rsc_pursuit_data.
 ↪parquet", engine="pyarrow")
#data_dir = Path("/Users/maygc/Documents/GitHub/pursuit")
#all_regions_data = pd.read_parquet("/Users/maygc/Documents/GitHub/pursuit/
 ↪ca1_ca3_rsc_pursuit_data.parquet", engine="pyarrow")
all_regions_data
```

[55]:
```
              time     ind  ratPos_1  ratPos_2  laserPos_1  laserPos_2  \
0         46.71875    2804   136.375  127.6875      153.25    115.5625
1            46.75    2805     136.5    127.25      153.25    114.9375
2         46.78125    2806   136.625    126.75      153.25    114.4375
3         46.78125    2807    136.75    126.25      153.25       114.0
4          46.8125    2808     137.0    125.75     153.375    113.5625
…              …       …         …         …           …           …
57147150    2086.0  125179  105.4375  48.34375    111.6875     66.5625
57147151    2086.0  125180   105.125   49.5625    111.9375       67.25
```

57147152	2086.0	125181	104.8125	50.78125	112.0625	67.8125
57147153	2086.0	125182	104.5625	52.0625	112.1875	68.4375
57147154	2086.0	125183	104.375	53.34375	112.8125	69.3125

	ratVel	ratAcc	laserVel	laserAcc	ratMoveDir	laserMoveDir \
0	28.453125	83.1875	30.0	27.328125	-1.301758	-1.327148
1	29.046875	83.4375	30.390625	30.21875	-1.257812	-1.319336
2	29.671875	83.0	30.8125	33.0625	-1.214844	-1.30957
3	30.328125	81.8125	31.25	35.8125	-1.172852	-1.298828
4	30.984375	79.875	31.71875	38.3125	-1.132812	-1.287109
…	…	…	…	…	…	…
57147150	72.6875	83.0	53.1875	107.0	1.803711	1.240234
57147151	72.25	86.4375	52.46875	101.75	1.786133	1.209961
57147152	71.6875	90.625	51.78125	96.5625	1.767578	1.179688
57147153	71.0	95.5	51.125	91.5	1.748047	1.151367
57147154	70.1875	100.9375	50.53125	86.6875	1.729492	1.124023

	ratHeadDir	ratAV	laserAV	laserDist	laserBearingMD \
0	-0.988281	-0.045044	0.008102	20.8125	0.676758
1	-0.981934	-0.044342	0.009621	20.78125	0.623535
2	-0.976074	-0.043243	0.010971	20.6875	0.577637
3	-0.969238	-0.041718	0.012146	20.5625	0.535156
4	-0.958984	-0.039825	0.0131	20.390625	0.492188
…	…	…	…	…	…
57147150	1.939453	0.017914	-0.030777	19.265625	-0.564453
57147151	1.925781	0.018188	-0.0298	18.9375	-0.582031
57147152	1.908203	0.018509	-0.028671	18.5	-0.598633
57147153	1.895508	0.01889	-0.027359	18.078125	-0.613281
57147154	1.885742	0.019348	-0.02594	18.046875	-0.646484

	laserBearingHD	centerDist	centerBearingMD	centerBearingHD \
0	0.363281	40.90625	-0.224365	-0.538086
1	0.348145	40.4375	-0.27002	-0.545898
2	0.339111	39.9375	-0.315918	-0.554199
3	0.331787	39.40625	-0.361084	-0.564941
4	0.318115	38.90625	-0.405762	-0.57959
…	…	…	…	…
57147150	-0.700195	35.3125	-0.80127	-0.937012
57147151	-0.72168	34.46875	-0.80957	-0.949707
57147152	-0.739746	33.625	-0.819336	-0.960449
57147153	-0.760254	32.75	-0.829102	-0.975586
57147154	-0.802734	31.84375	-0.839355	-0.995605

	spkTable_1	spkTable_2	spkTable_3	spkTable_4	spkTable_5 \
0	0	0	1	0	0
1	0	0	0	0	0
2	0	0	0	0	0

3	0	1	0	0	0
4	0	0	0	0	0
…	…	…	…	…	…
57147150	0	0	0	0	0
57147151	0	0	0	0	0
57147152	0	0	0	0	0
57147153	0	0	0	0	0
57147154	0	0	0	0	0

	spkTable_6	spkTable_7	spkTable_8	spkTable_9	spkTable_10 \
0	0	0	0	<NA>	<NA>
1	1	0	0	<NA>	<NA>
2	0	0	0	<NA>	<NA>
3	1	0	0	<NA>	<NA>
4	0	0	0	<NA>	<NA>
…	…	…	…	…	…
57147150	0	0	0	1	0
57147151	0	0	0	0	0
57147152	0	0	0	0	0
57147153	0	0	0	0	0
57147154	0	1	0	1	0

	spkTable_11	spkTable_12	spkTable_13	spkTable_14	spkTable_15 \
0	<NA>	<NA>	<NA>	<NA>	<NA>
1	<NA>	<NA>	<NA>	<NA>	<NA>
2	<NA>	<NA>	<NA>	<NA>	<NA>
3	<NA>	<NA>	<NA>	<NA>	<NA>
4	<NA>	<NA>	<NA>	<NA>	<NA>
…	…	…	…	…	…
57147150	0	0	0	0	0
57147151	0	0	0	0	0
57147152	0	0	0	0	0
57147153	0	0	0	0	0
57147154	0	0	0	0	1

	spkTable_16	spkTable_17	spkTable_18	spkTable_19	spkTable_20 \
0	<NA>	<NA>	<NA>	<NA>	<NA>
1	<NA>	<NA>	<NA>	<NA>	<NA>
2	<NA>	<NA>	<NA>	<NA>	<NA>
3	<NA>	<NA>	<NA>	<NA>	<NA>
4	<NA>	<NA>	<NA>	<NA>	<NA>
…	…	…	…	…	…
57147150	1	0	0	0	<NA>
57147151	0	0	0	0	<NA>
57147152	0	0	0	0	<NA>
57147153	0	0	0	0	<NA>
57147154	0	0	0	0	<NA>

```
          spkTable_21  spkTable_22  spkTable_23  spkTable_24  spkTable_25  \
0                <NA>         <NA>         <NA>         <NA>         <NA>
1                <NA>         <NA>         <NA>         <NA>         <NA>
2                <NA>         <NA>         <NA>         <NA>         <NA>
3                <NA>         <NA>         <NA>         <NA>         <NA>
4                <NA>         <NA>         <NA>         <NA>         <NA>
…                  …            …            …            …            …
57147150         <NA>         <NA>         <NA>         <NA>         <NA>
57147151         <NA>         <NA>         <NA>         <NA>         <NA>
57147152         <NA>         <NA>         <NA>         <NA>         <NA>
57147153         <NA>         <NA>         <NA>         <NA>         <NA>
57147154         <NA>         <NA>         <NA>         <NA>         <NA>

          spkTable_26  spkTable_27  spkTable_28  spkTable_29  spkTable_30  \
0                <NA>         <NA>         <NA>         <NA>         <NA>
1                <NA>         <NA>         <NA>         <NA>         <NA>
2                <NA>         <NA>         <NA>         <NA>         <NA>
3                <NA>         <NA>         <NA>         <NA>         <NA>
4                <NA>         <NA>         <NA>         <NA>         <NA>
…                  …            …            …            …            …
57147150         <NA>         <NA>         <NA>         <NA>         <NA>
57147151         <NA>         <NA>         <NA>         <NA>         <NA>
57147152         <NA>         <NA>         <NA>         <NA>         <NA>
57147153         <NA>         <NA>         <NA>         <NA>         <NA>
57147154         <NA>         <NA>         <NA>         <NA>         <NA>

          spkTable_31  spkTable_32  spkTable_33  spkTable_34 region  \
0                <NA>         <NA>         <NA>         <NA>    CA1
1                <NA>         <NA>         <NA>         <NA>    CA1
2                <NA>         <NA>         <NA>         <NA>    CA1
3                <NA>         <NA>         <NA>         <NA>    CA1
4                <NA>         <NA>         <NA>         <NA>    CA1
…                  …            …            …            …    …
57147150         <NA>         <NA>         <NA>         <NA>    CA1
57147151         <NA>         <NA>         <NA>         <NA>    CA1
57147152         <NA>         <NA>         <NA>         <NA>    CA1
57147153         <NA>         <NA>         <NA>         <NA>    CA1
57147154         <NA>         <NA>         <NA>         <NA>    CA1

         trial_block                 sessFile
0            pursuit  KB09_20_pursuitRoot.mat
1            pursuit  KB09_20_pursuitRoot.mat
2            pursuit  KB09_20_pursuitRoot.mat
3            pursuit  KB09_20_pursuitRoot.mat
4            pursuit  KB09_20_pursuitRoot.mat
…              …                        …
```

```
57147150     pursuit  LP06_47_pursuitRoot.mat
57147151     pursuit  LP06_47_pursuitRoot.mat
57147152     pursuit  LP06_47_pursuitRoot.mat
57147153     pursuit  LP06_47_pursuitRoot.mat
57147154     pursuit  LP06_47_pursuitRoot.mat

[57147155 rows x 58 columns]
```

[125]:
```
data_dir = Path("/Users/may/pursuit")
pursuit_data = pd.read_parquet("/Users/may/pursuit/all_regions_pursuit_indexed.
 ↪parquet", engine="pyarrow")
pursuit_data.head()
```

[125]:
```
     index      time   ind  ratPos_1   ratPos_2  laserPos_1  laserPos_2  \
268    268   51.1875  3072   120.375   31.03125    101.1875        73.5
269    269  51.21875  3073  120.4375   31.03125    101.4375        73.5
270    270  51.21875  3074  120.4375  31.015625    101.4375        73.5
271    271     51.25  3075     120.5       31.0    101.4375     73.9375
272    272  51.28125  3076     120.5  30.984375    101.4375     74.1875

       ratVel     ratAcc  laserVel  laserAcc  ratMoveDir  laserMoveDir  \
268  1.850586  14.804688     328.0    240.25   -0.661621      0.626465
269  2.082031  14.601562     323.0    431.25   -0.613281      0.625977
270  2.320312  14.320312     315.0     595.0   -0.581055      0.625488
271  2.556641  13.976562     304.0     730.0   -0.561035      0.624512
272  2.792969  13.578125     291.0     836.5   -0.550293      0.623535

     ratHeadDir     ratAV   laserAV  laserDist  laserBearingMD  \
268   -2.111328 -0.070068 -0.000636   46.59375         2.65625
269   -2.103516 -0.048248 -0.000751   46.50000        2.605469
270   -2.095703 -0.032166 -0.000899   46.53125        2.574219
271   -2.085938 -0.020142 -0.001074   47.00000        2.548828
272   -2.078125 -0.010979  -0.00127   47.21875        2.537109

     laserBearingHD  centerDist  centerBearingMD  centerBearingHD  spkTable_1  \
268       -2.175781    58.59375         1.922852        -2.910156           0
269         -2.1875    58.59375            1.875        -2.917969           0
270       -2.195312    58.59375          1.84375        -2.925781           0
271       -2.208984    58.59375         1.824219        -2.933594           0
272        -2.21875    58.59375         1.813477        -2.941406           0

     spkTable_2  spkTable_3  spkTable_4  spkTable_5  spkTable_6  spkTable_7  \
268           0           1           0           0           0           0
269           0           2           0           0           0           0
270           0           2           0           0           0           0
271           0           2           0           0           0           1
272           0           2           0           0           0           0
```

```
     spkTable_8  spkTable_9  spkTable_10  spkTable_11  spkTable_12  \
268           0        <NA>         <NA>         <NA>         <NA>
269           0        <NA>         <NA>         <NA>         <NA>
270           0        <NA>         <NA>         <NA>         <NA>
271           0        <NA>         <NA>         <NA>         <NA>
272           0        <NA>         <NA>         <NA>         <NA>

     spkTable_13  spkTable_14  spkTable_15  spkTable_16  spkTable_17  \
268         <NA>         <NA>         <NA>         <NA>         <NA>
269         <NA>         <NA>         <NA>         <NA>         <NA>
270         <NA>         <NA>         <NA>         <NA>         <NA>
271         <NA>         <NA>         <NA>         <NA>         <NA>
272         <NA>         <NA>         <NA>         <NA>         <NA>

     spkTable_18  spkTable_19  spkTable_20  spkTable_21  spkTable_22  \
268         <NA>         <NA>         <NA>         <NA>         <NA>
269         <NA>         <NA>         <NA>         <NA>         <NA>
270         <NA>         <NA>         <NA>         <NA>         <NA>
271         <NA>         <NA>         <NA>         <NA>         <NA>
272         <NA>         <NA>         <NA>         <NA>         <NA>

     spkTable_23  spkTable_24  spkTable_25  spkTable_26  spkTable_27  \
268         <NA>         <NA>         <NA>         <NA>         <NA>
269         <NA>         <NA>         <NA>         <NA>         <NA>
270         <NA>         <NA>         <NA>         <NA>         <NA>
271         <NA>         <NA>         <NA>         <NA>         <NA>
272         <NA>         <NA>         <NA>         <NA>         <NA>

     spkTable_28  spkTable_29  spkTable_30  spkTable_31  spkTable_32  \
268         <NA>         <NA>         <NA>         <NA>         <NA>
269         <NA>         <NA>         <NA>         <NA>         <NA>
270         <NA>         <NA>         <NA>         <NA>         <NA>
271         <NA>         <NA>         <NA>         <NA>         <NA>
272         <NA>         <NA>         <NA>         <NA>         <NA>

     spkTable_33  spkTable_34 region trial_block                 sessFile  \
268         <NA>         <NA>    CA1     pursuit  KB09_20_pursuitRoot.mat
269         <NA>         <NA>    CA1     pursuit  KB09_20_pursuitRoot.mat
270         <NA>         <NA>    CA1     pursuit  KB09_20_pursuitRoot.mat
271         <NA>         <NA>    CA1     pursuit  KB09_20_pursuitRoot.mat
272         <NA>         <NA>    CA1     pursuit  KB09_20_pursuitRoot.mat

     pursuit_task_id
268                3
269                3
270                3
```

```
271                 3
272                 3
```

[130]:
```python
#mass feature extraction time!

features_df = pursuit_functions.feature_extraction.
 ↪extract_laser_feats(pursuit_data)
features_df.head()
```

[130]:
```
   pursuit_task_id  curvature  path_efficiency  time_to_target  \
0                3   1.471040         0.588729          5.6875
1                4   5.933417         0.944167          1.9375
2                5   1.543211         0.689762          0.3750
3                6   2.030372         0.124926         10.1250
4                7   2.321145         0.923304          2.0000

   total_path_distance  path_ratio  avsq_laserHD  avsq_laserMD  \
0           151.750000    1.022241      2.570641      1.614854
1            44.156250    1.974522      0.350760      0.167632
2             9.804688    0.493625      3.226866      6.824451
3           362.500000    0.864138      2.274144      2.129579
4            75.500000    1.181291      1.443636      0.568921

   avsq_laserMoveDir
0           2.247310
1           3.787769
2           3.690575
3           3.706532
4           1.899775
```

[131]:
```python
features_df
```

[131]:
```
      pursuit_task_id  curvature  path_efficiency  time_to_target  \
0                   3   1.471040         0.588729          5.6875
1                   4   5.933417         0.944167          1.9375
2                   5   1.543211         0.689762          0.3750
3                   6   2.030372         0.124926         10.1250
4                   7   2.321145         0.923304          2.0000
…                 …         …                …               …
7696            13712   1.425981         0.980588          2.0000
7697            13713   2.089212         0.135455          7.0000
7698            13717   0.262923         0.364945          4.0000
7699            13718   0.074008         0.942839          2.0000
7700            13719   1.768937         0.862424          0.0000

      total_path_distance  path_ratio  avsq_laserHD  avsq_laserMD  \
0              151.750000    1.022241      2.570641      1.614854
```

```
1               44.156250    1.974522      0.350760      0.167632
2                9.804688    0.493625      3.226866      6.824451
3              362.500000    0.864138      2.274144      2.129579
4               75.500000    1.181291      1.443636      0.568921
…                     …           …             …             …
7696            61.406250    1.116539      0.498092      0.266060
7697           335.000000    0.920896      0.709909      0.600936
7698           173.750000    1.058993      0.648517      0.476314
7699           101.375000    0.100339      2.624615      0.107878
7700           141.500000    0.257509      2.106709      1.302491

      avsq_laserMoveDir
0              2.247310
1              3.787769
2              3.690575
3              3.706532
4              1.899775
…                   …
7696           3.636192
7697           2.919200
7698           5.217361
7699           2.750882
7700           3.332335

[7701 rows x 9 columns]
```

[135]:
```python
#PCA TIMEE!!!!

#standardize features
scaler = StandardScaler()
scaled_features = scaler.fit_transform(features_df)

#perform PCA
n_components = 2
pca= PCA(n_components=n_components)
pca_components = pca.fit_transform(scaled_features)
pca_df = pd.DataFrame(
    pca_components,
    columns=[f"PC{i+1}" for i in range(n_components)]
)

print(f'Explained variance ratio: {pca.explained_variance_ratio_}')
```

```
Explained variance ratio: [0.28774093 0.18486767]
```

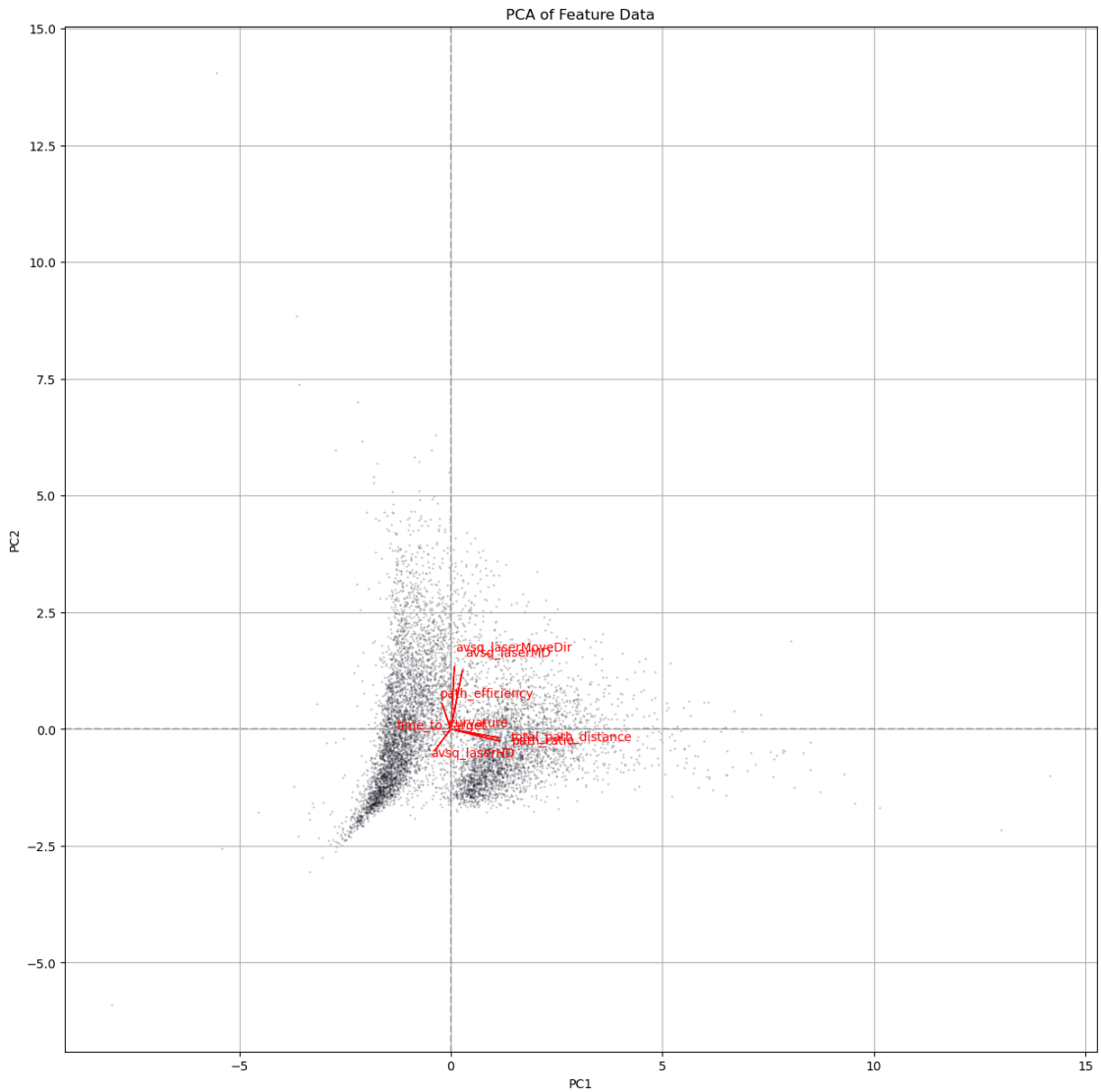
[138]:
```python
pca_loadings = pd.DataFrame(
    pca.components_,
    columns=features_df.columns,
```

```
    index=[f"PC{i+1}" for i in range(n_components)]
)
print(pca_loadings)

for i in range(n_components):
    top_features = pca_loadings.iloc[i].abs().nlargest(3)  # Get the top 3␣
 ↪features
    print(f"Top features for PC{i+1}:")
    print(top_features)
    print("\n")
```

```
     pursuit_task_id  curvature  path_efficiency  time_to_target  \
PC1        -0.025704  -0.105166        -0.515657        0.571803
PC2         0.020495   0.276307         0.000069       -0.094046

     total_path_distance  path_ratio  avsq_laserHD  avsq_laserMD  \
PC1             0.581214   -0.192473      0.136541      0.043076
PC2            -0.130572   -0.232352      0.623145      0.670272

     avsq_laserMoveDir
PC1           0.007991
PC2           0.076022
Top features for PC1:
total_path_distance    0.581214
time_to_target         0.571803
path_efficiency        0.515657
Name: PC1, dtype: float64


Top features for PC2:
avsq_laserMD    0.670272
avsq_laserHD    0.623145
curvature       0.276307
Name: PC2, dtype: float64
```

[147]:
```
#plot PCA

components = pca.components_
feature_names = features_df.columns[1:]

plt.figure(figsize=(15,15))

plt.scatter(pca_df["PC1"], pca_df["PC2"], c='blue', edgecolor='black', alpha=0.
 ↪5, s=0.1)
plt.xlabel("PC1")
```

```
plt.ylabel("PC2")
plt.title("PCA of Feature Data")

for i, feature in enumerate(feature_names):
    plt.arrow(0, 0, components[0, i] * 2, components[1, i] * 2, color='red',␣
 ↪alpha=1)
    plt.text(components[0, i] * 2.5, components[1, i] * 2.5, feature,␣
 ↪color='red')

plt.axhline(0, linestyle="dashed", color="grey", alpha=0.5)
plt.axvline(0, linestyle="dashed", color="grey", alpha=0.5)
plt.grid()
plt.show()
```



[231]:
```
# Extract trials from the first cluster (e.g., PC1 > 0, PC2 > 0)
cluster_1_trials = features_df[(features_df["PC1_x"] > 0) &␣
↪(features_df["PC1_x"] < 2) & (features_df["PC2_x"] < 0) &␣
↪(features_df["PC2_x"] > -0.5)]["pursuit_task_id"].tolist()

# Extract trials from the second cluster (e.g., PC1 < 0, PC2 < 0)
cluster_2_trials = features_df[(features_df["PC1_x"] < -2) &␣
↪(features_df["PC2_x"] < 0)]["pursuit_task_id"].tolist()

cluster_3_trials = features_df[(features_df["PC1_x"] > 0) &␣
↪(features_df["PC2_x"] > 0)]["pursuit_task_id"].tolist()

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# Print extracted trial IDs
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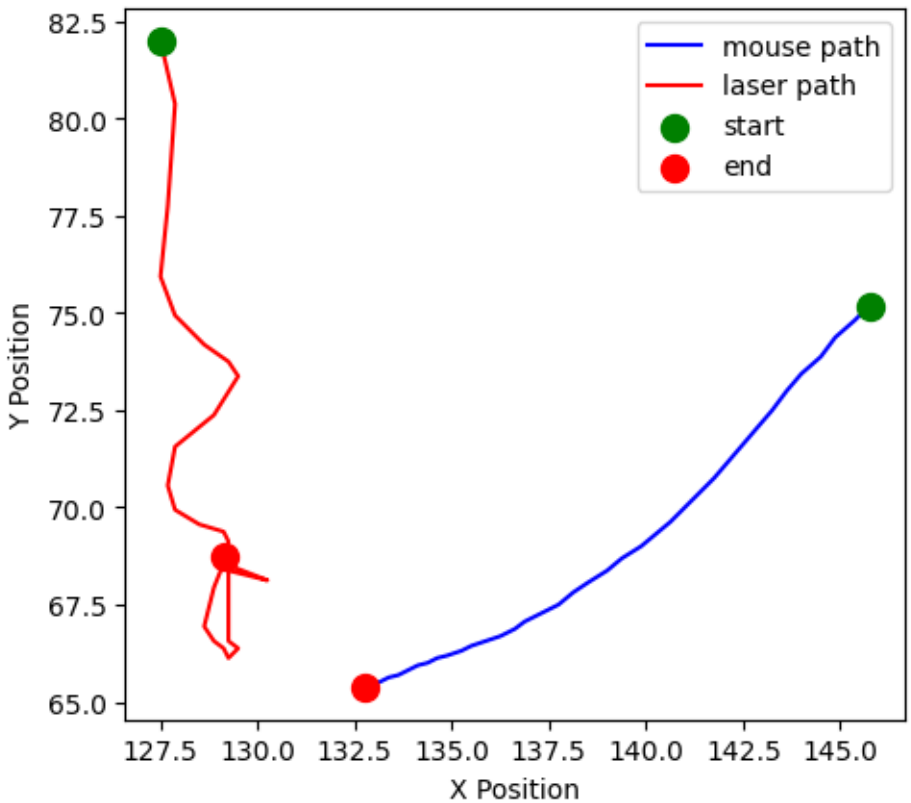
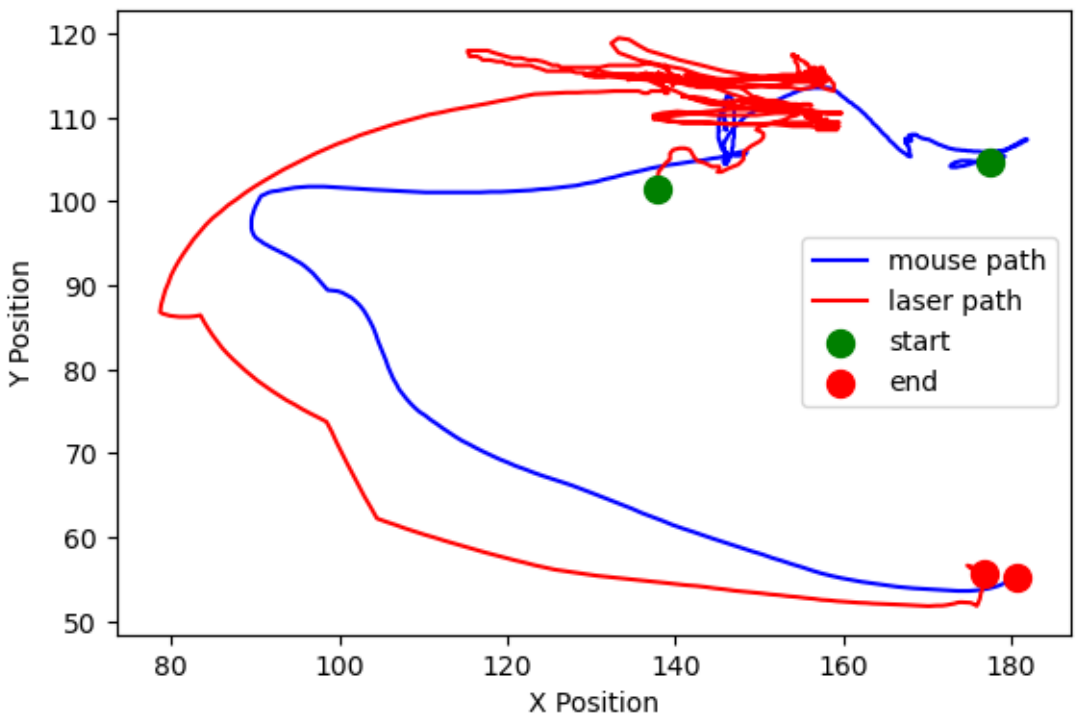
[273]:
```
#plot the other trials
lst = cluster_4_trials
n = np.random.choice(len(lst), 1, replace=False)
pursuit_functions.plot.
 ↪plot_trajectory(pursuit_data[pursuit_data['pursuit_task_id'] == lst[n[0]]])
```



[174]:
```
random_pursuits = np.random.choice(pursuit_data['pursuit_task_id'].unique(), 3,␣
↪replace=False)
print(random_pursuits)
# random_pursuits = np.array([13914, 61627,  7701])
pursuit_functions.plot.
↪plot_trajectory(pursuit_data[pursuit_data['pursuit_task_id'] ==␣
↪random_pursuits[0]])
pursuit_functions.plot.
↪plot_trajectory(pursuit_data[pursuit_data['pursuit_task_id'] ==␣
↪random_pursuits[1]])
pursuit_functions.plot.
↪plot_trajectory(pursuit_data[pursuit_data['pursuit_task_id'] ==␣
↪random_pursuits[2]])
```

```
[12773 11956  9690]
```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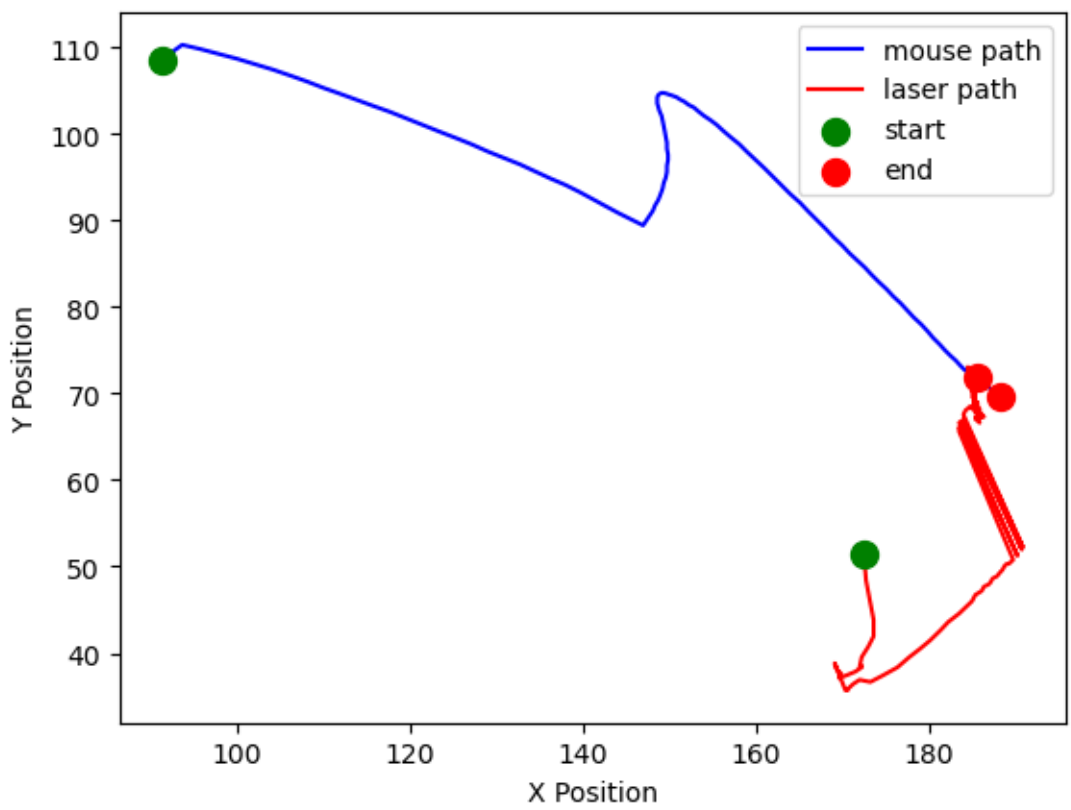
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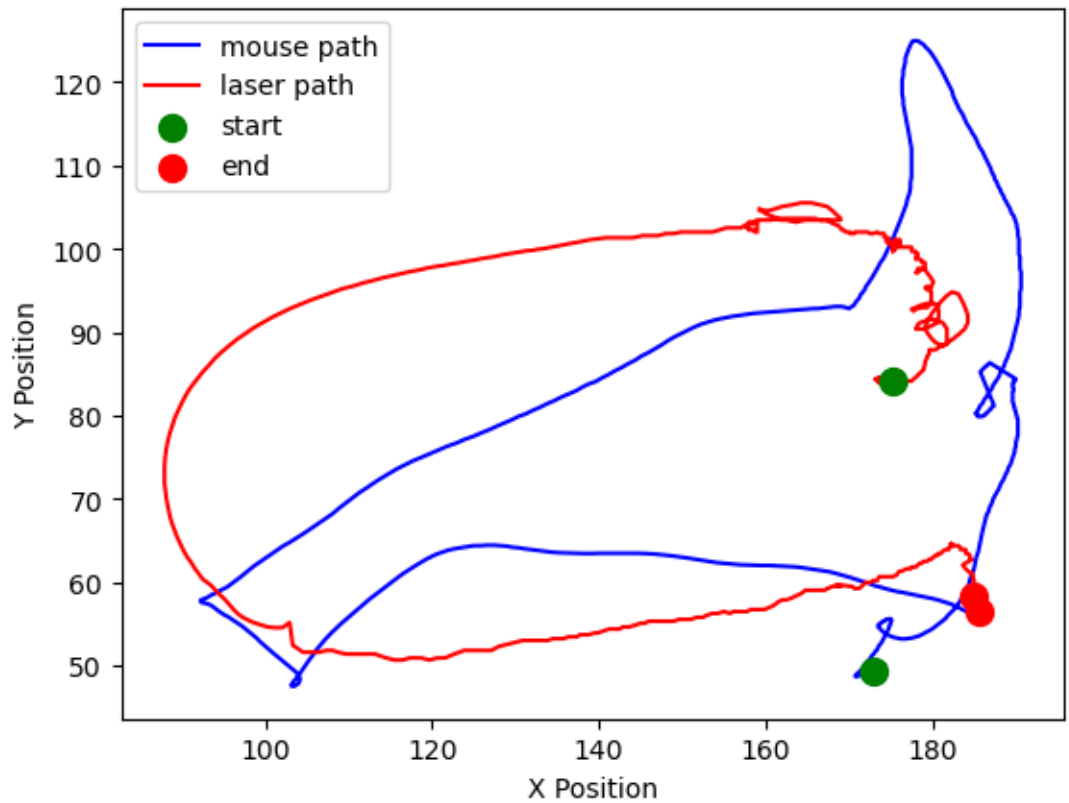

[220]:
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 63}
}
```

[177]:
```
print(trial_types)
```

```
{'Pseudorandom': {6786, 1699, 4104, 8908, 2092, 3725, 11437, 13650, 2323, 4988,
2781}, 'Characteristic': {8064, 6882, 10018, 12773, 10311, 4012, 11342, 6959,
5968, 2225, 11735, 3992, 9690, 6427, 4926}}
```

[221]:
```
#adding trial_type to features_df
features_df["trial_type"] = features_df["pursuit_task_id"].apply(
    lambda x: next((key for key, ids in trial_types.items() if x in ids),␣
 ↪"Unknown")
)
features_df.head()
```

[221]:
```
   pursuit_task_id  curvature  path_efficiency  time_to_target  \
0                3   1.471040         0.588729          5.6875
1                4   5.933417         0.944167          1.9375
2                5   1.543211         0.689762          0.3750
3                6   2.030372         0.124926         10.1250
4                7   2.321145         0.923304          2.0000

   total_path_distance  path_ratio  avsq_laserHD  avsq_laserMD  \
0           151.750000    1.022241      2.570641      1.614854
1            44.156250    1.974522      0.350760      0.167632
2             9.804688    0.493625      3.226866      6.824451
3           362.500000    0.864138      2.274144      2.129579
4            75.500000    1.181291      1.443636      0.568921

   avsq_laserMoveDir trial_type     PC1_x     PC2_x     PC1_y     PC2_y
0           2.247310    Unknown -0.168431 -0.106580 -0.168431 -0.106580
1           3.787769    Unknown -1.748008 -1.649728 -1.748008 -1.649728
2           3.690575    Unknown -0.731931  3.020219 -0.731931  3.020219
3           3.706532    Unknown  1.414691  0.056849  1.414691  0.056849
4           1.899775    Unknown -1.330967 -1.048755 -1.330967 -1.048755
```

[214]:
```
#merge pca_df to features_df
features_df = features_df.merge(pca_df[['PC1', 'PC2']], left_index=True,␣
 ↪right_index=True)
features_df.head()
```

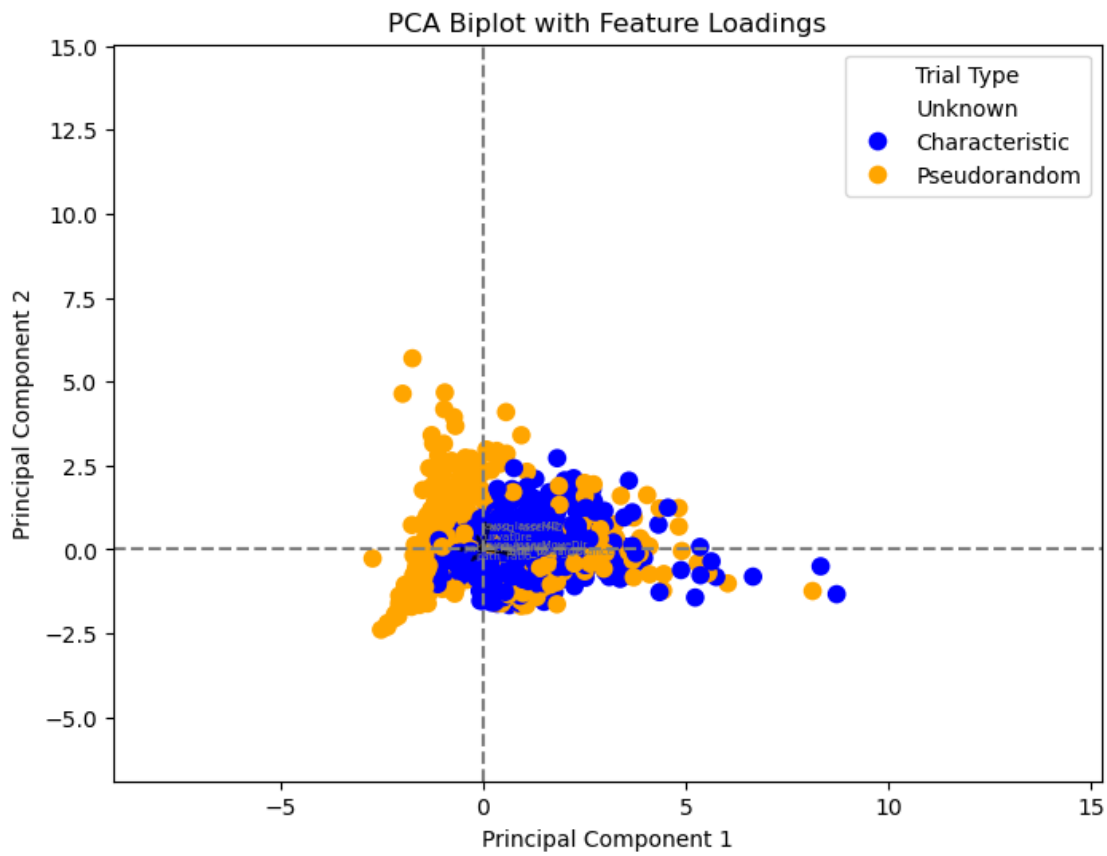
[214]:
```
   pursuit_task_id  curvature  path_efficiency  time_to_target  \
0                3   1.471040         0.588729          5.6875
1                4   5.933417         0.944167          1.9375
2                5   1.543211         0.689762          0.3750
3                6   2.030372         0.124926         10.1250
4                7   2.321145         0.923304          2.0000
```

```
   total_path_distance  path_ratio  avsq_laserHD  avsq_laserMD  \
0           151.750000    1.022241      2.570641      1.614854
1            44.156250    1.974522      0.350760      0.167632
2             9.804688    0.493625      3.226866      6.824451
3           362.500000    0.864138      2.274144      2.129579
4            75.500000    1.181291      1.443636      0.568921

   avsq_laserMoveDir trial_type     PC1_x     PC2_x     PC1_y     PC2_y
0           2.247310    Unknown -0.168431 -0.106580 -0.168431 -0.106580
1           3.787769    Unknown -1.748008 -1.649728 -1.748008 -1.649728
2           3.690575    Unknown -0.731931  3.020219 -0.731931  3.020219
3           3.706532    Unknown  1.414691  0.056849  1.414691  0.056849
4           1.899775    Unknown -1.330967 -1.048755 -1.330967 -1.048755
```

[193]:
```
pca_loadings = pd.DataFrame(
    pca.components_.T,
    index=[col for col in features_df.columns if col not in ["PC1", "PC2",␣
 ↪"trial_type"]],  # Exclude PC columns
    columns=[f"PC{i+1}" for i in range(pca.n_components_)]
)
```

[223]:
```
#plot PCA with trial labels
custom_palette = {
    "Characteristic": "blue",
    "Pseudorandom": "orange",
    "Unknown": (0,0,0,0),
}

plt.figure(figsize=(8,6))
sns.scatterplot(
    data=features_df,
    x="PC1_x", y="PC2_x",
    hue="trial_type",  # Color points by trial type
    palette=custom_palette,
    edgecolor = "none",
    # alpha=0.7,
    s=70
)

for feature in pca_loadings.index:
    plt.arrow(0, 0,
              pca_loadings.loc[feature, "PC1"],
              pca_loadings.loc[feature, "PC2"],
              color="black", alpha=0.6, head_width=0.1)
    plt.text(pca_loadings.loc[feature, "PC1"],
             pca_loadings.loc[feature, "PC2"],
```

```
            feature, fontsize=5, color="gray")

plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.title("PCA Biplot with Feature Loadings")
plt.axhline(0, color="grey", linestyle="dashed")
plt.axvline(0, color="grey", linestyle="dashed")
plt.legend(title="Trial Type")
plt.show()
```

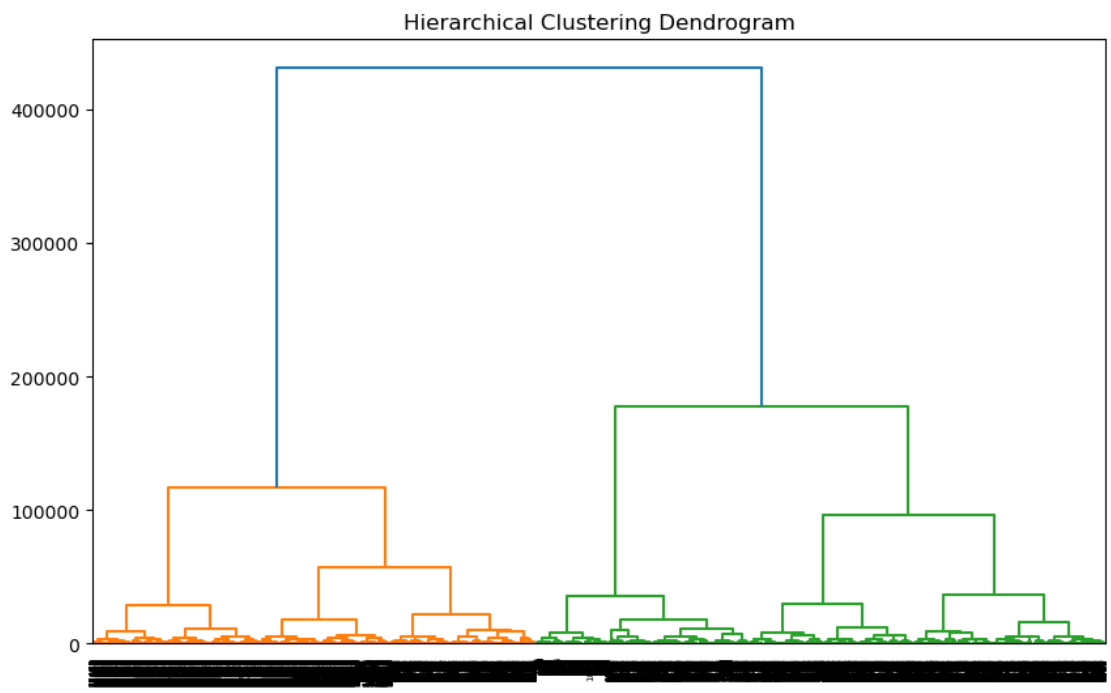


[225]:
```
features_df[(features_df['PC1_x'] > 6) & (features_df['trial_type'] ==␣
↪"Characteristic")]
```

[225]:
```
      pursuit_task_id  curvature  path_efficiency  time_to_target  \
1013             1699   3.849113         0.102235           65.00
1833             3111   1.821291         0.136606           43.00
4410             8021   1.728686         0.042371           42.75

      total_path_distance  path_ratio  avsq_laserHD  avsq_laserMD  \
1013               1275.0    0.713333      2.829805      2.928447
```

```
1833                1402.0    0.434379      1.786342      2.650604
4410                2272.0    0.596391      2.016581      2.412954

      avsq_laserMoveDir      trial_type     PC1_x     PC2_x     PC1_y  \
1013           3.697953  Characteristic  8.321946 -0.507256  8.321946
1833           3.327762  Characteristic  6.652850 -0.815152  6.652850
4410           2.992672  Characteristic  8.716824 -1.337891  8.716824

         PC2_y
1013 -0.507256
1833 -0.815152
4410 -1.337891
```

[278]:
```
from scipy.cluster.hierarchy import linkage, dendrogram
features_numeric = features_df.select_dtypes(include=[np.number])

# Compute hierarchical clustering
linkage_matrix = linkage(features_numeric, method="ward")

# Plot dendrogram
plt.figure(figsize=(10, 6))
dendrogram(linkage_matrix, labels=features_df["pursuit_task_id"].values,␣
 ↪leaf_rotation=90)
plt.title("Hierarchical Clustering Dendrogram")
plt.show()
```



[307]:
```
from scipy.cluster.hierarchy import fcluster

# Choose a threshold to cut the tree into 2 clusters (adjust based on your␣
↪dendrogram)
threshold = 0.25 * max(linkage_matrix[:, 2])  # 70% of the max height
cluster_labels = fcluster(linkage_matrix, threshold, criterion='distance')

# Add cluster labels to the DataFrame
features_df["cluster"] = cluster_labels
```

[309]:
```
features_df["cluster"].unique()
```

[309]:
```
array([3, 4, 2, 1], dtype=int32)
```

[310]:
```
cluster_1_trials_HC = features_df.loc[features_df["cluster"] == 1,␣
↪"pursuit_task_id"].tolist()
cluster_2_trials_HC = features_df.loc[features_df["cluster"] == 2,␣
↪"pursuit_task_id"].tolist()
cluster_3_trials_HC = features_df.loc[features_df["cluster"] == 3,␣
↪"pursuit_task_id"].tolist()
cluster_4_trials_HC = features_df.loc[features_df["cluster"] == 4,␣
↪"pursuit_task_id"].tolist()

print("Cluster 1 Trials HC:", cluster_1_trials)
print("Cluster 2 Trials HC:", cluster_2_trials)
```

```
Cluster 1 Trials HC: [7743, 7839, 7845, 7848, 7850, 7852, 7854, 7859, 7862,
7866, 7868, 7872, 7874, 7876, 7883, 7887, 7889, 7895, 7898, 7913, 7916, 7918,
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7935, 7942]
```
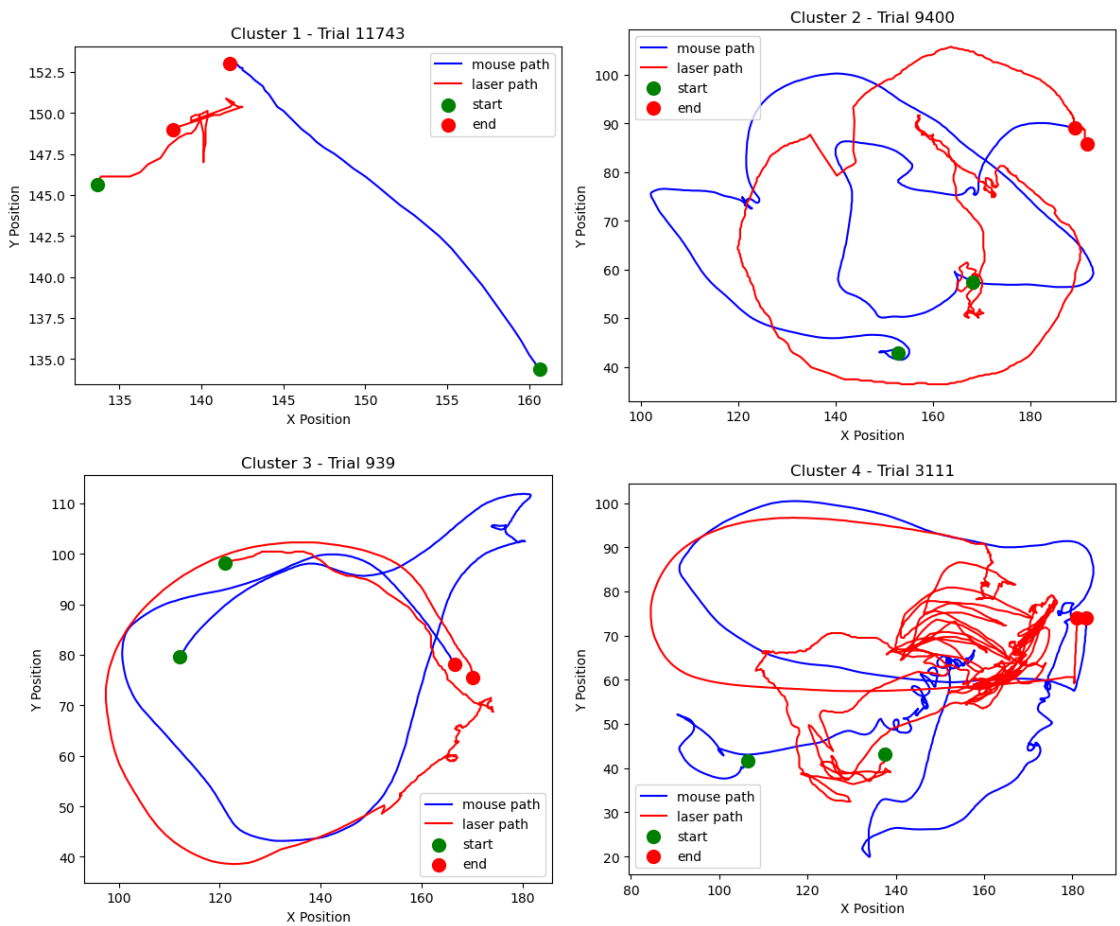
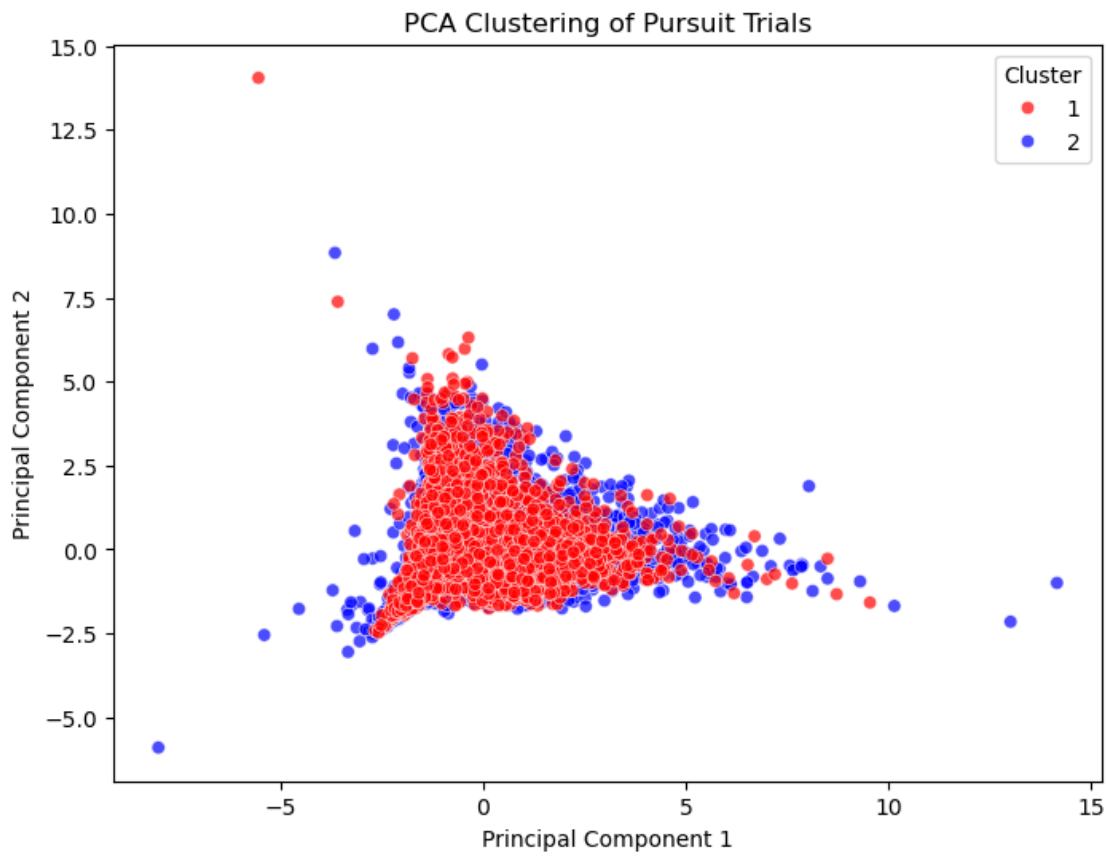

[324]:

```python
#plot the other trials
import matplotlib.pyplot as plt
import numpy as np

# Define your lists of trials
lists_of_trials = {
    "Cluster 1": cluster_1_trials_HC,
    "Cluster 2": cluster_2_trials_HC,
    "Cluster 3": cluster_3_trials_HC,
    "Cluster 4": cluster_4_trials_HC
}

# Create a 2x2 subplot grid
fig, axes = plt.subplots(2, 2, figsize=(12, 10))
axes = axes.flatten()  # Flatten to make indexing easier

# Loop through each list and plot a random trajectory
for i, (cluster_name, trial_list) in enumerate(lists_of_trials.items()):
    if len(trial_list) > 0:
        # Select a random trial ID from the list
        trial_id = np.random.choice(trial_list, 1)[0]

        # Plot trajectory in the corresponding subplot
        pursuit_functions.plot.
 ↪plot_trajectory(pursuit_data[pursuit_data['pursuit_task_id'] == trial_id],␣
 ↪ax=axes[i])
```

```
        # Set title with cluster name
        axes[i].set_title(f"{cluster_name} - Trial {trial_id}")

# Adjust layout for better spacing
plt.tight_layout()
plt.show()
```



[281]:
```
plt.figure(figsize=(8,6))
sns.scatterplot(
    data=features_df,
    x="PC1_x", y="PC2_x",
    hue="cluster",  # Color by cluster instead of trial type
    palette=["red", "blue"],  # Change colors for each cluster
    alpha=0.7
)
plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.title("PCA Clustering of Pursuit Trials")
```

```
plt.legend(title="Cluster")
plt.show()
```